How Much Targeted Investment Does It Take to Top a Multi-Task Leaderboard? It Depends on the Aggregation Rule

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Abstract

A developer who wants a model to top a multi-task leaderboard can fold data related to some sub-datasets into its training, raising its scores there. We ask how large this *targeted investment* must be for a given target model to rank first, and how this depends on the rule that aggregates sub-dataset scores. We define investment as the minimum cost of raising the target's scores, within a per-task cap, so that it is strictly first while all other models stay fixed, and we measure it as the number of tasks touched (K), the total score points (L_1), and the largest single-task raise. We solve this exactly (closed forms and MILPs, tested against brute force) for nine aggregation rules on two public leaderboards: the Open LLM Leaderboard (6 tasks, about 4500 models) and HELM MMLU (57 subjects, 79 models). The answer is conditional. Under a permissive cap (scores may rise to 100) the arithmetic mean is the most expensive rule in points; for the twenty closest challengers on the Open LLM Leaderboard the median costs 3.6 points [95% CI 1.3, 9.0] against 29.2 [26.4, 31.4] for the mean, and on HELM the Condorcet-winner requirement costs 0.30 [0.28, 0.31] of the mean's points, yet Borda and the Condorcet requirement touch many more of the 57 subjects. Under a cap limiting the raise the picture changes: with a cap anchored in observed fine-tuning gains, first place becomes infeasible beyond a rule-dependent rank on the Open LLM Leaderboard (for the mean, beyond rank 110; far earlier for min, median and Condorcet) and the mean is the most feasible rule there; on HELM with a fixed $+10$-point cap plurality and median are the most feasible rules, whereas a cap transferred from the Open LLM Leaderboard leaves 85–96% of HELM targets infeasible for every rule. Under the permissive cap, L_1 grows with current rank with log-log slopes of 0.24–0.64 on the Open LLM Leaderboard. The rule ordering is not robust to score normalisation or to dropping one task; we report where it holds and where it fails.

1 Introduction

A multi-task benchmark scores every model on several sub-datasets and aggregates the scores into one ranking, most often by averaging. Rankings influence which models are adopted, which papers are published and where funding goes, so developers have reason to act strategically. One such action is *targeted investment*: folding data related to some sub-datasets into a model's training to raise its scores on those sub-datasets. Benchmark leakage and contamination are known to inflate scores (Zhou et al., 2023; Yang et al., 2023), and strategic practices distort leaderboards (Singh et al., 2025), but it is less clear *how much* improvement is needed to change a ranking, and whether the answer depends on how the sub-dataset scores are aggregated.

We study this for a single target model that wants to rank first, with all other models held fixed. Our question is: what investment makes the target strictly first, and how does it vary with the aggregation rule? Because "investment" has no canonical measure we report three (tasks touched K, score points L_1, largest single-task raise L_∞) and a convex variant, and because the answer depends on how far a score can plausibly be raised we make the per-task *cap* an explicit modelling assumption and vary it. Infeasibility, where no plan within the caps reaches first place, is an outcome we report rather than drop.

Contributions.

- **A precise formulation** (Section 3) of minimum targeted investment under score-based and rank-based aggregation rules, with exact solvers for nine rules (closed forms or water-filling for score rules, MILPs for rank rules) validated against brute force, and for the mean, median and min against closed-form propositions on every target (Section 4.2).
- **An empirical comparison on public data** (Open LLM Leaderboard, 6 tasks; HELM MMLU, 57 subjects) of how K, L_1 and feasibility depend on the rule, the target's rank and the cap (Sections 4.3–4.5), with bootstrap intervals over targets.
- **Conditions of validity** (Section 4.6): which conclusions survive raw versus normalised scores, leaderboard slices, task granularity, evaluation noise and leaving one task out, and which do not. Two predictions we made in advance (that the L_1 of median and min is of the same order as the mean's or larger, and that "concentrate on one task" is the cheapest plan for typical targets) are contradicted or narrowed by the data.

The answer we give is therefore conditional: investment is a distribution over rank, rule, cap and cost, and the claims below always carry the cap they were measured under. We do not model rival reactions, we do not train any model, and the caps are scenario assumptions rather than measurements of what contamination can achieve (Section 5).

2 Related work

Our statements about cited works rest on their abstracts and bibliographic records, not on a full reading; the Handbook of Computational Social Choice and the evaluation harness are cited as general references.

Fragility of multi-task benchmarks. Dehghani et al. (2021) show, on several common benchmark setups, that the relative performance of methods can change substantially with the choice of benchmark tasks (the "benchmark lottery"). Their perturbation is the choice of tasks by the evaluator; ours is a targeted change of one model's scores, and our question is quantitative: how much change is needed, under which rule. Lanctot et al. (2023) view each task as a voter and bring social-choice rules to cross-task evaluation of agents and language models; we use their viewpoint to select rank-based rules (Borda, Condorcet winner, plurality) next to score-based rules (mean, median, minimum, geometric, harmonic, power and trimmed means), and we compare how costly it is to win under each. Standard references on these voting rules are collected in the Handbook of Computational Social Choice (Brandt et al., 2016).

Strategic behaviour and contamination. Singh et al. (2025) document practices on the Chatbot Arena leaderboard (private testing of many variants, selective score disclosure, unequal sampling) that distort rankings. Benchmark leakage and contamination, i.e. data related to an evaluation set being used for training, is studied empirically by Zhou et al. (2023), who report that it can substantially boost scores, and Yang et al. (2023), who show that rephrased test samples evade common decontamination and let a 13B model overfit a benchmark. These works establish that targeted improvement of a benchmark's score is feasible and consequential; they do not ask what the *minimum* improvement for first place is, nor how it depends on how sub-dataset scores are aggregated. Our cap scenarios (Section 3) are a modelling stand-in for what such data can achieve, not a measurement of it. The closest listed work, Chen et al. (2026), models benchmarking as a Stackelberg game between a benchmark designer and several developers who can inflate observed scores by allocating resources to benchmark-specific improvements, and gives equilibrium results. We differ in setting and method: a single target against fixed rivals, exact minimum-cost solutions per aggregation rule, and empirical evaluation on public leaderboard data.

Minimum-cost manipulation in voting. Computational social choice studies the cost of changing an election's input so that a chosen candidate wins, e.g. bribery, where an external actor pays voters to change their preferences (Faliszewski et al., 2009). Minimum investment to rank first is the analogue for score matrices: the "voters" are tasks, the "bribe" is a raise of the target's score on a task, and the cost is measured in score points or number of tasks touched. The setting differs in that

the manipulated quantity is a cardinal score bounded by a headroom cap, and the instances are real leaderboards with n of order 6–57 tasks.

Evaluation data. We use the HELM MMLU release (Liang et al., 2022) over the 57 MMLU subjects (Hendrycks et al., 2021), and the Open LLM Leaderboard (tasks IFEval, BBH (Suzgun et al., 2022), MATH Lvl 5, GPQA, MuSR, MMLU-Pro), whose tasks are run with the Language Model Evaluation Harness (Gao et al., 2023).

3 Setting and method

Leaderboard and aggregation rules. A leaderboard is a score matrix $X \in \mathbb{R}^{M\times n}$ (higher is better) of M models on n sub-datasets ("tasks"), and an aggregation rule R that maps X to a ranking of the models. Rules are either *score rules*, which compute a scalar $f(x_j)$ from each model's score vector (mean, geometric, harmonic and power means with $p < 1$, median, minimum, trimmed mean), or *rank rules*, which depend only on per-task orderings (Borda/mean rank, plurality as first-place count, and the requirement of being a Condorcet winner). Being a Condorcet winner is sufficient for first place under every Condorcet-consistent rule (Copeland, Schulze, Kemeny, ranked pairs), so its minimal investment is an upper bound on theirs.

Targeted investment. A developer wants a *target* model i to rank first. Training on data related to task t lets them raise the target's score on t from x_{it} to any $y_t \in [x_{it}, c_t]$, where c_t is a cap (headroom); all other models are held fixed. The target is *first* if it is strictly above every other model under R (margin $\varepsilon = 10^{-3}$ score points). A cap scenario is the central modelling assumption: we study $c_t = 100$ (ceil), the best existing score on t plus 0 or 2 points (front), a fixed gain of $5/10/20$ points over the target's own score, and empirically anchored caps equal to the 95th/99th percentile of fine-tune-minus-base differences measured on the Open LLM Leaderboard. *Minimal investment* is $\min_y \mathrm{cost}(y - x_i)$ over feasible y making i first. We report several costs side by side: K, the number of tasks touched (breadth); $L_1 = \sum_t (y_t - x_{it})$ (volume); $L_\infty = \max_t (y_t - x_{it})$; $L_2 = \sum_t (y_t - x_{it})^2$ (a convex cost under which concentrating on one task is no longer free); and u^*, the smallest uniform gain δ on every task that makes i first, the "general improvement" reference. If no feasible y exists the target is *infeasible*; this is an outcome that we report as a fraction and never drop or impute.

Exact solvers. For each rule we compute minimal investment exactly rather than by heuristic search. For the mean, $L_1^* = n(\tau + \varepsilon) - \sum_t x_{it}$ in closed form, where τ is the best rival's value, and K^* follows by raising the tasks with the most headroom first. For geometric, harmonic and power means ($p < 1$, concave) the L_1 optimum equalises the shifted scores at a water level, found by bisection; K is a greedy selection. For median and min the optimum is an order-statistic argument (how many tasks must reach the rivals' decisive score). The trimmed mean and the rank rules are solved as mixed-integer programs with HiGHS: a raise on task t to just above a rival's score passes that rival on t (a tie counts one half), so the candidate levels are finite, and rival constraints are generated lazily. Being the Condorcet winner is the requirement that the target beat every rival in a majority of tasks; since it is sufficient for first place under every Condorcet-consistent rule, we also solve the Copeland rule exactly on HELM to see how tight this bound is. Every solver is tested against brute-force enumeration on small random instances, and the closed forms for the mean, median and min are compared against the solver output on every target (Section 4.2). A solve that reaches its 120 s limit is flagged and never recorded as infeasible; none did in any file that records the flag (Appendix A).

4 Experiments

4.1 Setup

Data. **Open LLM Leaderboard (OLLB)**: the public `contents` snapshot (4576 rows; 4497 models after removing duplicate names, 4496 after excluding one flagged model) with six tasks (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro) in the leaderboard's own normalised scale. **HELM MMLU**: release v1.13.0, 79 models on 57 subjects (accuracy). Versions, commits

and checksums are in Appendix A. Because OLLB scores are rescaled per task (e.g. GPQA tops out near 29), scale-sensitive rules depend on normalisation, so we also run the raw scores (Section 4.6).

Targets, rules, caps. Every model of rank ≥ 2 under the mean is a target: 4495 on OLLB and 78 on HELM. Score rules (mean, geometric, harmonic, median, min) run on all targets. Borda, Condorcet-winner, plurality and the trimmed mean are exact but expensive on OLLB, so they run on a rank-stratified sample of 57 OLLB targets (ranks 2–21 plus a log-uniform sample of the rest, seed 0); on HELM every rule runs on every target. Cross-rule comparisons therefore use either the 20 closest challengers (ranks 2–21), on which every rule was run, or ratios paired per target. A target of mean-rule rank ≥ 2 can already be first under another rule (at most one per rule and dataset: HELM geometric, harmonic, min and plurality; OLLB median and min; one Borda target in the sample); its K and L_1 are 0 and it is *included* in Table 1, Tables 3–6, Figure 1 and the slopes, but excluded from the uniform-plan ratios (Figure 4). We present caps c_t as defined in Section 3; the headline cap is the empirical one (`emp95`), the 95th percentile of the gain a fine-tuned model shows over its base model on that task, measured on 540 base/fine-tune pairs in OLLB, and `ceil` is the permissive bound. HELM has no base/fine-tune pairs, so its `emp95` cap is the OLLB MMLU-Pro quantile (about 2.9 raw points) transferred as a uniform per-subject cap; every HELM statement under that cap is conditional on this transfer. Margin $\varepsilon = 10^{-3}$ points.

Uncertainty. Group medians carry 95% percentile bootstrap intervals over targets (1000 resamples, seed 0); feasibility fractions carry Wilson intervals; medians of K and L_1 are over feasible targets and say so. Targets in neighbouring ranks are not independent (many OLLB entries are merges or fine-tunes of the same base), so intervals describe sampling of targets from this leaderboard, not of leaderboards; we add leaderboard slices below.

4.2 Theory anchor: closed forms agree with the solvers

For the mean, $L_1^* = n(\tau + \varepsilon) - \sum_t x_{it}$ matches the solver on every feasible row to 10^{-11} points or better; for median and min the closed-form predictions for K (and L_1 where derived) match on every row where a prediction exists (Figure 5 in Appendix D: 15 666 mean, 4865 median and 17 550 min rows for K, all equal). Two of our prior expectations are narrowed or contradicted by the data. (i) The median needs at least $\lfloor n/2 \rfloor + 1$ tasks *only in the worst case*, when the target is below the rivals' decisive score on every task. Among the 20 closest challengers under the ceiling the median needs $K = 1$ on OLLB ($n = 6$) and 10 [6.5, 10.5] on HELM ($n = 57$), the min needs 2 and 3.5 [2, 5]: these challengers already beat the decisive scores on most tasks. (ii) The median and the min do not cost as many points as the mean, but *fewer*: the paired ratio $L_1(\text{median})/L_1(\text{mean})$ is 0.60 [0.60, 0.60] on OLLB and 0.156 [0.14, 0.19] on HELM, and for min 0.251 [0.248, 0.253] and 0.147 [0.12, 0.18] (ceiling, all targets). The mean must buy the whole gap to the leader times n in points; order-statistic rules only fix the few decisive tasks. "Raise only one task" is likewise a property of the head of a 6-task leaderboard: under the ceiling the mean needs $K = 1$ for the 20 closest OLLB challengers, but only 9.2% of all OLLB targets, and $K = 7$ [4.5, 9] on HELM.

4.3 Rule ordering of the minimum investment

Under the permissive cap the mean is the most expensive rule in points. For the 20 closest OLLB challengers (ranks 2–21, ceiling), the median L_1 is: median rule 3.6 [1.3, 9.0], Borda 6.3 [4.8, 8.8], min 10.4 [6.3, 15.2], Condorcet winner 11.4 [10.4, 14.4], plurality 16.4 [14.7, 19.1], harmonic 18.0 [13.3, 20.8], geometric 21.9 [18.8, 23.8], trimmed mean 24.4 [21.3, 26.0], mean 29.2 [26.4, 31.4]. On HELM the order is plurality 16.0 [6.5, 21.1], min 17.0 [6.9, 25.3] and median 17.8 [11.4, 34.0] (intervals overlap), then Condorcet 64.8 [24.5, 94.1], Borda 147 [55, 206], harmonic 242, geometric 258 and mean 278 [177, 360]. Paired on the same targets, the ratio of each rule's L_1 to the mean's (ceiling, all targets) is, on HELM, 0.961 [0.953, 0.967] (geometric), 0.925 (harmonic), 0.776 [0.725, 0.823] (Borda), 0.295 [0.282, 0.307] (Condorcet), 0.156 (median), 0.147 (min) and 0.068 [0.063, 0.073] (plurality); on OLLB 0.852 (geometric), 0.697 (harmonic), 0.599 (median), 0.251 (min), and for the 57-target sample 0.372 [0.333, 0.436] (Borda), 0.640 [0.560, 0.655] (Condorcet), 0.500 [0.447, 0.572] (plurality) and 0.815 (trimmed mean). Table 1 gives the headline-cap version and Figure 1 shows how L_1 and feasibility depend on the target's rank.

Table 1: Minimum investment under the headline cap (`emp95`) for the 20 closest challengers (ranks 2–21) and all other targets: median tasks touched K and points L_1 [95% bootstrap CI] over feasible targets, and number of infeasible targets out of all (a target already first under the rule counts with $K = L_1 = 0$). HELM uses the cap transferred from OLLB; † rules run on the 57-target OLLB sample (trimmed mean has no K); single numbers have fewer than five feasible targets.

	ranks 2–21			rank >21		
rule	K	L_1	infeas.	K	L_1	infeas.
Open LLM Leaderboard (emp95)						
mean	2.0	29.2 [26.5, 31.2]	0/20	4.0	66.2 [65.0, 68.1]	4160/4475
geometric mean	4.0	35.6 [21.6, 39.9]	2/20	5.0	65.2 [60.8, 69.7]	4382/4475
harmonic mean	3.0	19.8 [2.0, 45.2]	9/20	4.0	57.0 [47.8, 62.6]	4409/4475
median	1.0	3.3 [1.0, 8.5]	2/20	2.0	20.7 [18.7, 22.3]	4352/4475
min	1.0	3.1 [0.89, 6.2]	13/20	2.0	6.0 [4.8, 7.2]	4417/4475
Borda†	2.0	6.3 [4.9, 9.0]	0/20	3.0	22.2 [12.4, 32.2]	14/37
Condorcet winner†	3.0	10.9 [8.3, 12.5]	8/20	4.0	18.0	34/37
plurality†	2.0	16.7 [15.4, 19.6]	4/20	2.0	27.5 [19.6, 33.4]	24/37
trimmed mean†	–	24.4 [21.3, 26.0]	0/20	–	27.5 [26.1, 30.1]	24/37
HELM MMLU (emp95, transferred)						
mean	15.0	41.8 [4.0, 156]	15/20	–	–	58/58
geometric mean	10.0	28.6	16/20	–	–	58/58
harmonic mean	4.0	10.9	17/20	–	–	58/58
median	3.5	2.4 [0.38, 9.9]	12/20	–	–	58/58
min	1.5	2.2	16/20	–	–	58/58
Borda	28.0	45.2 [5.7, 59.5]	13/20	–	–	58/58
Condorcet winner	17.0	16.5 [0.89, 24.5]	13/20	–	–	58/58
plurality	8.0	6.5 [2.5, 17.1]	8/20	–	–	58/58

Where this ordering fails or is unresolved. (i) For rank 2–5 there are only four targets per dataset; intervals straddle 1 (for example OLLB ceiling Condorcet 0.91 [0.41, 1.41], plurality 0.84 [0.59, 1.09]), so we make no ordering claim there. (ii) Under the empirical cap the geometric mean is not cheaper than the mean where both are feasible (1.05 [1.00, 1.07], 111 of 4495 targets), and on HELM at a fixed $+10$ cap geometric and harmonic means are indistinguishable from it (0.99 [0.97, 1.04], 0.999 [0.93, 1.03]). (iii) *Breadth is a different statement from volume*: on HELM, Borda needs 28.5 [27, 29] *more* tasks than the mean, Condorcet 12 [9, 14] more, median 4 [3, 4] more, while min needs 4 [3, 4] fewer and plurality 7 [4, 9.5] fewer; on OLLB the differences are 0–2 tasks. Borda and Condorcet are cheaper in points but spread the raise over many of the 57 subjects. (iv) The ordering depends on normalisation and on the task set (Section 4.6).

Investment grows with rank. Under the ceiling, the log-log slope of L_1 against the target's mean-rule rank (feasible targets of rank ≥ 2, bootstrap CI) is 0.44 [0.43, 0.44] (mean) on OLLB, with 0.51 [0.50, 0.52] (geometric), 0.64 [0.62, 0.67] (median), 0.61 [0.58, 0.63] (min), 0.60 [0.54, 0.66] (Borda), 0.44 [0.42, 0.46] (Condorcet) and 0.24 [0.20, 0.29] (plurality); on HELM it is 1.13 [0.91, 1.32] (mean), 1.07 (geometric), 2.17 [1.63, 2.76] (median), 1.76 [1.17, 2.41] (Condorcet), 1.67 [1.48, 1.89] (min), 1.48 [1.25, 1.80] (Borda) and 1.52 [1.12, 2.08] (plurality). Under capped scenarios the slopes are fitted only to the feasible survivors, which biases them towards flatter values, so we do not interpret them as elasticities.

4.4 Caps decide who can reach first place at all

Under the ceiling, front $(+0, +2)$ caps no target on either leaderboard is infeasible for any rule. Under the empirical cap the situation reverses (Table 2, Figure 2): on OLLB, first place under the mean is infeasible for 92.5% of targets (nearly all with rank > 100; the smallest infeasible rank is 110), whereas min, Condorcet and harmonic have infeasible targets already at ranks 5–6 and median at 13. *The ordering of rules by cost flips with the cap*: the mean is the most expensive rule in points under the ceiling, but under `emp95` it is the most feasible rule beyond rank 20, because the compensatory mean can spread a small gain over six tasks while min, median and the Condorcet requirement need specific tasks lifted past specific rivals. This is dataset-dependent: on HELM under

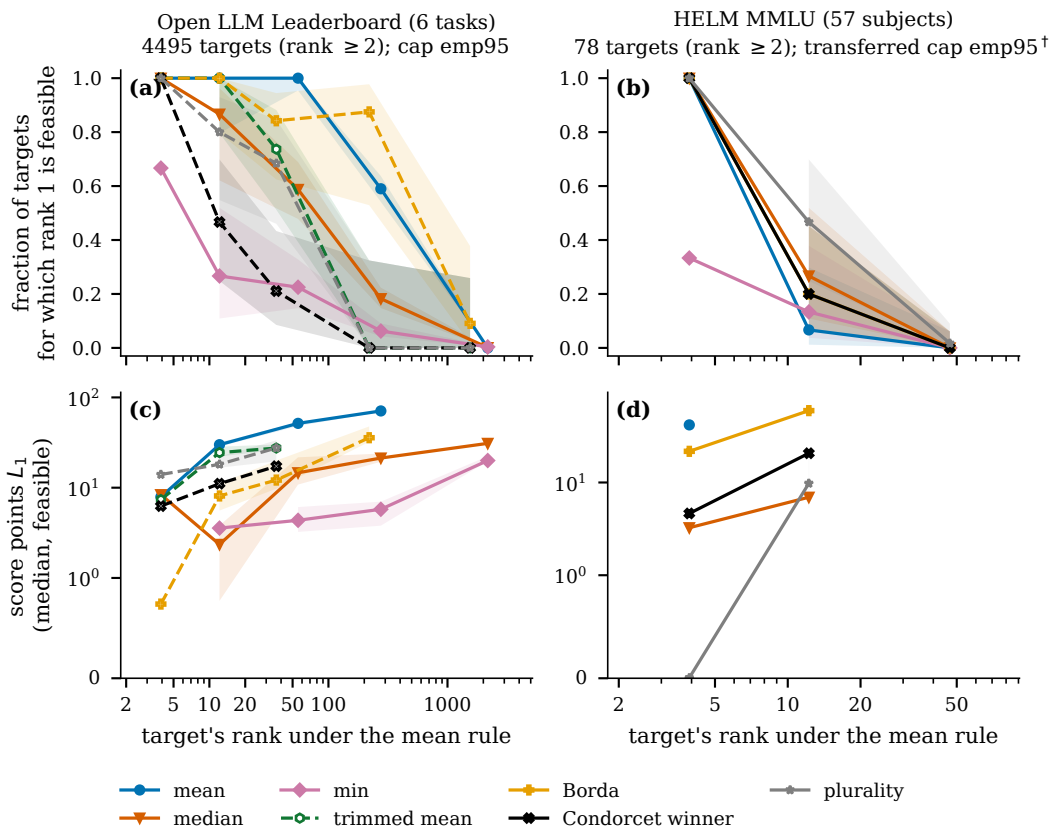


Figure 1: Investment against the target's current rank under the mean rule, headline cap (OLLB `emp95`; HELM `emp95` transferred from OLLB†). Top: fraction of targets for which first place is feasible. Bottom: median score points L_1 over feasible targets per rank bin (the median K is in Table 1) (bands: 95% bootstrap CI where at least five targets are feasible). Dashed lines: rules run on the 57-target OLLB sample. HELM bins hold 4, 15 and 59 targets. †HELM has no base/fine-tune pairs.

Table 2: Share of targets for which first place is infeasible, OLLB, cap `emp95`, by the target's mean-rule rank. Score rules: 4/15/80/4396 targets per bin; * rules run on the 57-target sample: 4/15/19/19.

rule	2–5	6–20	21–100	>100	smallest infeasible rank
mean	0	0	0	0.946	110
Borda*	0	0	0.158	0.579	22
trimmed mean*	0	0	0.263	1	37
plurality*	0	0.200	0.316	1	8
median	0	0.133	0.412	0.982	13
geometric	0	0.133	0.575	0.986	11
harmonic	0.250	0.533	0.662	0.991	5
min	0.250	0.733	0.775	0.991	5
Condorcet*	0	0.533	0.789	1	6

a +10 cap plurality (27% infeasible) and median (49%) are the most feasible rules and min the least (82%); under the transferred `emp95` 85–96% of targets are infeasible for every rule, and the smallest infeasible rank is 3 (min) to 10 (median). So "any model can reach first place" (ceiling) becomes "only models within roughly the top 100 (OLLB, mean) or top 6 (HELM, mean) can" under the empirical caps. The fractions for Borda, trimmed mean, plurality and Condorcet on OLLB come from 4/15/19/19 targets per rank bin and are correspondingly uncertain (with 19 targets a share of 0.16 has a 95% Wilson interval of about 0.05–0.38).

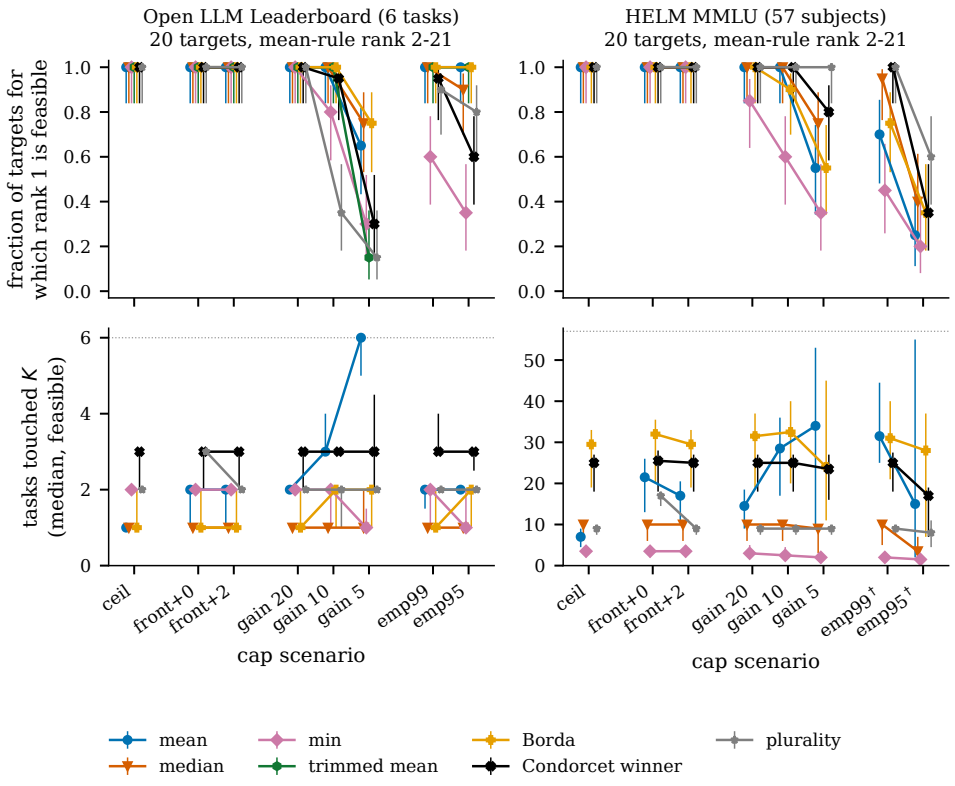


Figure 2: Cap sensitivity for the 20 closest challengers (mean-rule rank 2–21), the only set on which every rule was run on every target. Top: fraction feasible (95% Wilson). Bottom: median K over feasible targets (95% bootstrap CI); its composition changes as feasibility drops. Cap families (ceiling, rival front, fixed gain, empirical) are not ordered against each other. †HELM empirical caps are transferred from OLLB.

4.5 Targeted versus uniform improvement

The uniform plan raises every task by the same amount u^*; it is the min-max (L_∞) plan, so $L_\infty = u^*$ in every row (maximum deviation 10^{-6}). The comparison is therefore the volume ratio $L_1(\text{targeted})/L_1(\text{uniform}) \leq 1$. For the mean it is exactly 1 in every group, dataset and cap: any plan reaching the same total costs the same, targeting only concentrates the points (OLLB: $K = 1$ against 6 tasks; HELM ceiling, rank 6–20: $K = 8$ [7, 9] against 57). For every other rule targeting saves points: OLLB, ceiling, rank 6–20: min 0.27 [0.21, 0.30], plurality 0.25 [0.22, 0.30], median 0.33, Borda 0.36 [0.34, 0.39], Condorcet 0.37 [0.33, 0.41], harmonic 0.44 [0.43, 0.45], geometric 0.60 [0.58, 0.64], trimmed mean 0.67; HELM: min 0.044, median 0.111, plurality 0.111, Condorcet 0.32, Borda 0.60, harmonic 0.76, geometric 0.87 [0.86, 0.88] (Figure 4, Appendix D). Targeting is close to uniform (more than half of targets at ratio ≥ 0.9) only for the geometric mean under tight caps (OLLB `emp95` 0.95 [0.89, 0.97]; HELM $+10$ 0.93 [0.91, 0.96]) and, on HELM $+10$, the harmonic mean (0.87, 40% of targets). Rank 2–5 groups have three or four targets and wide intervals.

4.6 Robustness and conditions of validity

We summarise the stability of the rule ordering by Kendall's τ between the order of rules by median L_1 (20 closest challengers) in a variant and in the main run (Figure 3a,b). **Normalisation.** On raw OLLB scores the ordering changes substantially ($\tau = 0.21$ under the ceiling, 0.43 under a $+10$ cap): the median rule moves from cheapest (3.6) to 16.5 [14.6, 17.9], the mean from 29.2 to 19.0 [15.4, 19.7], and the arithmetic mean is no longer the most expensive rule; Borda stays among the cheapest. **Leaderboard slices.** Removing merged models ($\tau = 1.0$ under the ceiling, 0.79 under

$+10$) or keeping the top 500 (0.86, 0.64) changes head-20 medians by at most 2.8 points (no merges) or, in the top-500 slice, by up to 7.4 points (plurality 9.0 against 16.4), and swaps are between rules whose intervals overlap; the ordering of Section 4.3 holds on these slices. **Leaving one task out** (OLLB, challengers redefined on the remaining five tasks): τ against the full six-task ordering, ceiling, is 0.93 (BBH), 0.79 (IFEval), 0.71 (MATH Lvl 5), 0.57 (GPQA), 0.43 (MMLU-Pro) and -0.07 for MuSR; dropping MuSR reorders the rules almost completely, so the six-task ordering depends on which task the challengers are strong on. **Number of tasks.** On HELM, in random task subsets of size $n \in \{3, 6, 12, 24, 48\}$ (50 draws each, $+10$ cap, targets of rank 5 and 2 within each subset), the median L_1 of the rank-5 challenger grows roughly linearly in n for the mean (log-log slope 1.06) and the geometric mean (1.02), with K slopes 0.94 and 0.92, while for median and min L_1 is flat in n (-0.07, -0.08) and K stays between 1 and 2.5; Borda, Condorcet and plurality have L_1 slopes 0.89, 0.62 and -0.17. Slopes for the rank-2 challenger are noisy (some negative) and we make no claim of a scaling law from two ranks and no CI on these slopes. **Noise.** With evaluation noise (binomial standard error per cell, 100 draws, ten closest challengers, $+10$ cap) the coefficient of variation of L_1 is 0.17 [0.13, 0.53] (HELM) and 0.20 [0.18, 0.48] (OLLB) for the mean but 0.68 and 0.65 for the median and 0.75 (44% of draws infeasible) and 0.20 (51% infeasible) for min; under a task bootstrap on HELM it is 0.22 for the mean and 0.92 for min. The rules with the smallest L_1 (median, min) are the least stable, so their advantage in Section 4.3 is smaller than the sampling variation of a single target's cost.

4.7 Further checks

Details are in Appendix C. The Condorcet-winner bound is tight within ε against exact Copeland on HELM (not tested on OLLB). Under a convex (quadratic) cost the optimal plan touches all tasks, so "concentrate on one task" is an artefact of a linear cost. Random weights change the mean's cost by a factor of about two, and a tie margin of 0.1 instead of 10^{-3} changes median L_1 by at most 3.6% but can shift HELM K by a task or two.

5 Discussion and limitations

What the evidence says. "How much investment is needed" has no single answer: it is a distribution over the target's rank, the aggregation rule, the cap and the cost measure. Three statements hold, each with its conditions; the ordering of rules by cost is the least robust of them. (1) Under the permissive cap (ceiling) and normalised scores, the mean-like rules (arithmetic, geometric, harmonic) cost the most points. The cheapest rules are median, Borda and min on OLLB (3.6, 6.3, 10.4 points for the 20 closest challengers; plurality 16.4 is above them) and plurality, min and median on HELM; Borda and the Condorcet requirement lie in between and are strongly dataset-dependent (Section 4.3). This ordering does *not* survive every check: on raw OLLB scores the mean is no longer the most expensive rule ($\tau = 0.21$ against the main ordering), and without MuSR the ordering is reshuffled almost completely ($\tau = -0.07$); it holds on the no-merge and top-500 slices (Section 4.6). (2) Under a cap that limits the raise (fixed $+10$ points, or the empirical cap from fine-tuning gains), whether first place is reachable at all depends on rank and rule, and on OLLB the mean is the *most* reachable rule while min, median and Condorcet are the least; on HELM the order differs (Section 4.4). A rule that is cheapest to win under one cap is therefore not the hardest to win under another, and no rule is uniformly "harder to game" in our data. (3) Targeting a few tasks saves points over a uniform improvement under every rule except the mean, where it only concentrates the same points (Section 4.5). The contradicted expectation is instructive: order-statistic rules do not demand more points than the mean, only specific tasks, and their cost for a single target is the least stable under noise.

Where the conclusions fail. The ordering of rules fails under raw scores and without MuSR (above). With four targets in ranks 2–5 we say nothing about the closest challengers beyond "often indistinguishable from the mean". Where the cap binds, medians are over survivors and slopes are biased flat.

Limitations. *Model.* One target acts alone; rivals do not react, so results are minimum costs under a static leaderboard, not predictions of developer behaviour or equilibria. Raising a score on one task is assumed to cost nothing on others and the score is assumed controllable to any value up to the cap.

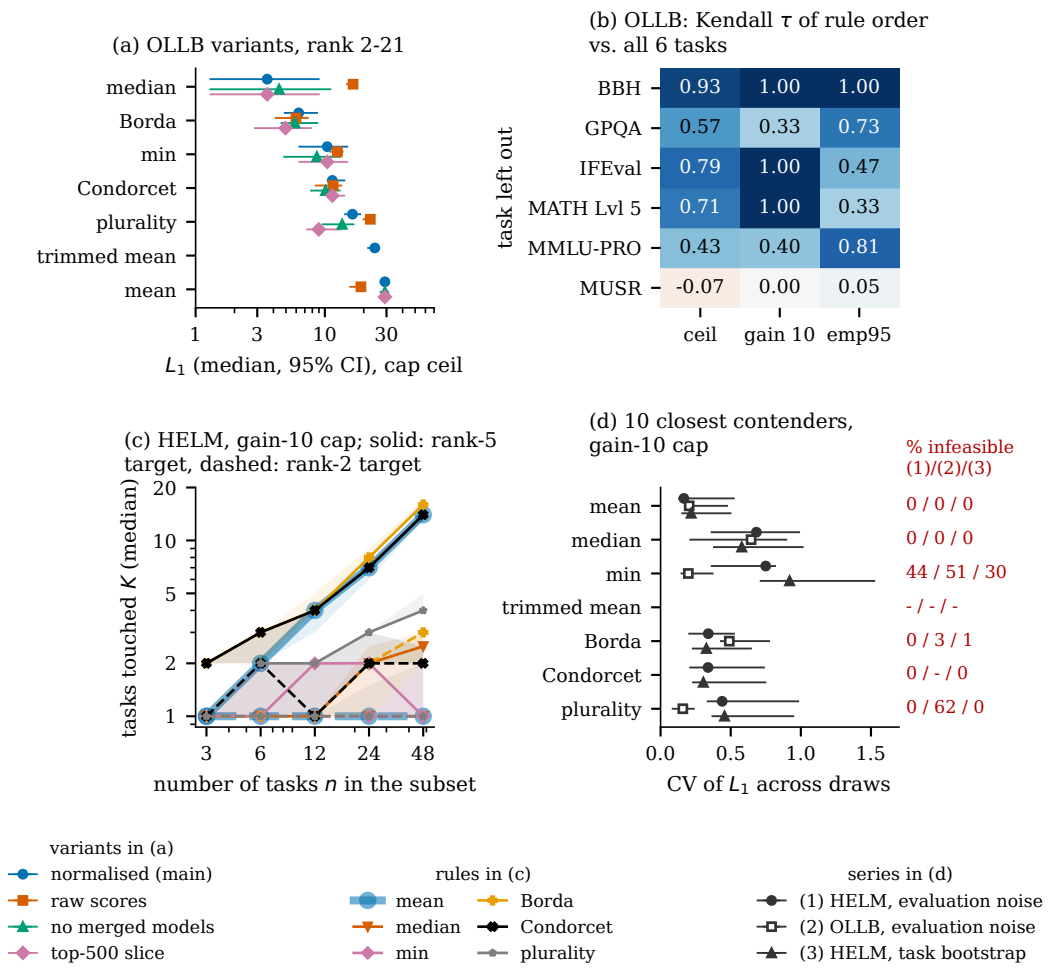


Figure 3: Robustness. (a) Median L_1 (95% CI) for the 20 closest OLLB challengers under the ceiling, for normalised scores (main), raw scores, without merged models and in the top-500 slice. (b) Kendall τ of the rule order with one task left out against all six tasks. (c) HELM: median K against the number of tasks in random subsets (+10 cap; solid rank-5, dashed rank-2 target). (d) Coefficient of variation of L_1 over draws under evaluation noise and task bootstrap (ten closest challengers); red column: percentage of infeasible draws for the numbered series (1), (2), (3) of the legend.

Caps. The caps are the central assumption. Our empirical cap is the 95th/99th percentile of observed fine-tune-minus-base gains on 540 OLLB pairs; this is an indirect proxy for what contamination or targeted data could achieve, not a measurement of it. HELM has no such pairs, so its empirical cap is a transfer of one OLLB task's quantile. *Data.* Two leaderboards of one benchmark family, one of them with only six tasks (so median and trimmed rules are coarse), and many OLLB entries are merges or fine-tunes of one another; intervals are over targets, not over leaderboards. *Coverage.* On OLLB, Borda, Condorcet, plurality and the trimmed mean were solved on a 57-target sample, the trimmed mean has no K, Copeland was solved only on HELM, the convex cost only for mean and geometric mean, and weighted means only for the mean. The robustness draws use ten (noise, task bootstrap) or two (task subsets) challengers and a single cap. *Compute.* The experiments exceed the task's 4 CPU-hour estimate; MILP time-limit flags are not recorded for the robustness draws (Appendix A).

Outlook. Extensions: reacting rivals (cf. Chen et al. 2026), caps measured from contamination experiments, leaderboards with more tasks.

Reproducibility statement

All numbers, figures and tables are produced by the code in the supplementary material from the public score tables pinned in Appendix A (commit hashes and SHA-256 checksums, cleaning steps, seeds, commands, expected run times, and the script that produces each figure and table); the solvers are tested against brute force and the closed forms against the solver output on every target. The full computation exceeds the compute estimate given with the task and takes several CPU-hours on two CPUs. ICLR 2027 has no mandatory checklist; the official NeurIPS and ICML 2026 files could not be fetched, so no checklist of those venues is included (the AI use statement is required by ICLR, the reproducibility and ethics statements are recommended).

Ethics statement

The study analyses published evaluation scores and trains no model. It quantifies how cheaply a leaderboard ranking can be moved under assumed caps, which could in principle inform strategic behaviour, but it relies on public data, ignores any particular developer's practices, and is intended to inform benchmark design and interpretation. The two score tables carry no declared licence; we redistribute derived costs and small processed score matrices only (Appendix A).

AI use statement

This study, including the literature search, analysis code, experiments, figures, tables and the text of this paper, was carried out by cooperating generative-AI agents (Claude models) working in separate roles (planning, analysis, presentation, writing, review). The task and its constraints were given by the operators of the pipeline. The agents were used for problem formulation, for writing and testing all code, for running all computations and for drafting and revising all text. No generative AI was used to produce data: every number in the paper was computed by the accompanying code from the public leaderboard snapshots listed in Appendix A. Code was checked by unit tests against brute-force solutions and by independent review agents; every cited reference was looked up by an agent in a bibliographic source (for several, the abstract only, as stated in Section 2), not independently verified by a human. The responsibility for the final content rests with the operators of the pipeline.

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A Reproduction details

The complete instructions are in `analysis/REPRODUCE.md` of the supplementary material; this appendix gives the information the paper's claims rest on.

Environment. Python 3.11.2, numpy 2.4.6, scipy 1.17.1 (HiGHS MILP, 120 s limit per solve, 300 s for Copeland), pandas 3.0.6, pyarrow 26.0.0, pytest 9.2.0; Linux, two CPUs and 4 GB per container, no GPU. Figures use Matplotlib and the paper TeX Live (pdflatex, bibtex).

Data and versions.

- Open LLM Leaderboard `contents`: Hugging Face dataset `open-llm-leaderboard/contents`, commit `9c09a7cae43334062a82cb164f2ef2550139dafa2` (2025-03-20), file `data/train-00000-of-00001.parquet`, SHA-256 `f13a7061e63133bc8745d89f8c7a56148584940ede22d3b27e21cad1e078dca5`, 4576 rows; 79 duplicate names removed (4497), one flagged model excluded (4496); no-merge variant 3789 models.
- Open LLM Leaderboard `results`: commit `aa81ecc38fdc5708254b833923368970efdf5ef5` (2025-03-15); used only to read the number of items per task (noise model, normalisation floors).
- HELM MMLU, release v1.13.0 (2025-01-10), 57 subjects $\times$ 79 models; per-file hashes in the data manifest (hash of the manifest `830bdb10a5518c50689d60c86623cc3d79ff5317d7e6b96408ea7dc2058de12b`). No other version was used.

Hashes of the processed matrices are in `analysis/data/processed/manifest.json`.

Licences. Neither score table declares a licence (HELM: the code is Apache-2.0, the bucket and site state nothing for the scores; OLLB `contents`: no licence field, a per-model `Hub License` column; `results`: no dataset card). We use published evaluation scores, redistribute only derived cost tables and small processed score matrices needed to run the code, and do not commit raw downloads. Formal clearance would have to come from the data owners.

Seeds and determinism. Seed 0 throughout (target sample, evaluation-noise, task-bootstrap and task-subset draws, Dirichlet weights, bootstrap intervals with 1000 resamples). Closed forms and MILPs are deterministic.

Commands and run time. `./reproduce.sh data`, `test`, then one target per experiment, then `summaries` (see `REPRODUCE.md` for the list). Individual steps take from under a minute to about 40 minutes on two CPUs; the sum is several CPU-hours (roughly 3–5 h sequentially, not measured as one run), *more than the 4 CPU-hours estimated in the task*. Then `figures/build_all.sh` (about 2 minutes) and `paper/build.sh`.

Disclosures. (i) The rank-rule L_∞ and uniform-gain columns were first computed with a tie convention that gave a tie at a rival's score no credit; a consistency check showed $L_\infty = \infty$ with L_1 finite on some `front0` plurality rows. It was fixed, covered by a test against brute-force min-max, and every file containing rank-rule rows was regenerated; all reported numbers come from the regenerated files. (ii) A mid-run edit of `reproduce.sh` made a few background jobs exit with an error; they were rerun. (iii) The robustness-draw files were produced before the time-limit flag existed; they contain no missing values, and we independently verified the trimmed-mean feasibility labels, but not those of the draw files.

Withheld paper. The task forbids searching for or reading the published paper that studied this question, or any page citing it. In the work behind this paper no such page was knowingly searched for, opened or encountered: literature searches covered the starter literature (read through the arXiv metadata API) and component topics (the bribery-in-elections literature, through OpenAlex metadata), and data and code searches stayed on the dataset locations given with the task and package documentation. No encounter was reported to us by any role.

B SCRIPT-TO-OUTPUT MAP

All inputs are result files of `analysis/results/`; every figure and table has a provenance file listing the SHA-256 of each input. Figure and table scripts are in `figures/scripts/`, result tables in `analysis/results/tables/`.

output	script	claim
Table 1, Tables 3–6	`table1_rule_dataset.py`	rule ordering, feasibility
Table 2	`feasibility_by_rank.csv` (`summarize_extra.py`)	feasibility
Fig. 1, 6	`fig1_investment_vs_rank.py`	rule ordering, rank, caps
Fig. 2	`fig2_cap_sensitivity.py`	caps
Fig. 4	`fig3_targeted_vs_uniform.py`	targeted vs uniform
Fig. 3	`fig4_robustness.py`	robustness
Fig. 5	`fig5_theory_check.py`	closed forms
Numbers quoted in text only	`results/tables/*.csv` (`FINDINGS.md` lists file and column)	all

Numbers quoted in the text and not shown in a figure or table are read from the summary tables listed in `analysis/FINDINGS.md`, which gives the file and column of each.

C FURTHER CHECKS

Condorcet versus Copeland (HELM, exact MILP, 7 caps $\times$ 78 targets). The Condorcet-winner bound is tight within ε: median paired L_1 ratio Copeland/Condorcet is 1 [1, 1] in every cap and rank group except `front0` (0.984–0.987, at most 1.3 points cheaper), the bound is never violated, and Copeland never reaches first place where the Condorcet requirement cannot. It is untested on OLLB, where the Copeland MILP over 4496 models was not run. **Convex cost.** Under a quadratic cost the optimal plan touches all tasks ($K = 6$ on OLLB, 57 on HELM for the mean and geometric mean), whereas the L_1 optimum touches 1–8 tasks for head targets under the ceiling: "concentrate on one task" is an artefact of a linear cost. Only the mean and the geometric mean were run with this cost. **Weighted mean** (200 seeded Dirichlet weight vectors): the median per-target ratio of the weighted to the equal-weight L_1 is 0.475 [0.473, 0.479] (OLLB, ceiling) and 0.400 [0.375, 0.423] (HELM), with a wide 5–95% range (0.21–0.85 on OLLB); the ranking of challengers is almost unchanged (Spearman 0.997 and 1.0). Weighting therefore changes the cost by a factor of about two, but the direction is confounded with a change of rivals. **Tie margin.** With $\varepsilon = 0.1$ instead of 10^{-3}, median L_1 changes by 0 to $+3.6\%$ and infeasible shares by at most 2.6 percentage points; K is less stable

on HELM (the share of targets whose K changes reaches 86% for the geometric mean under $+10$), so we do not quote HELM K to the unit.

D ADDITIONAL RESULTS

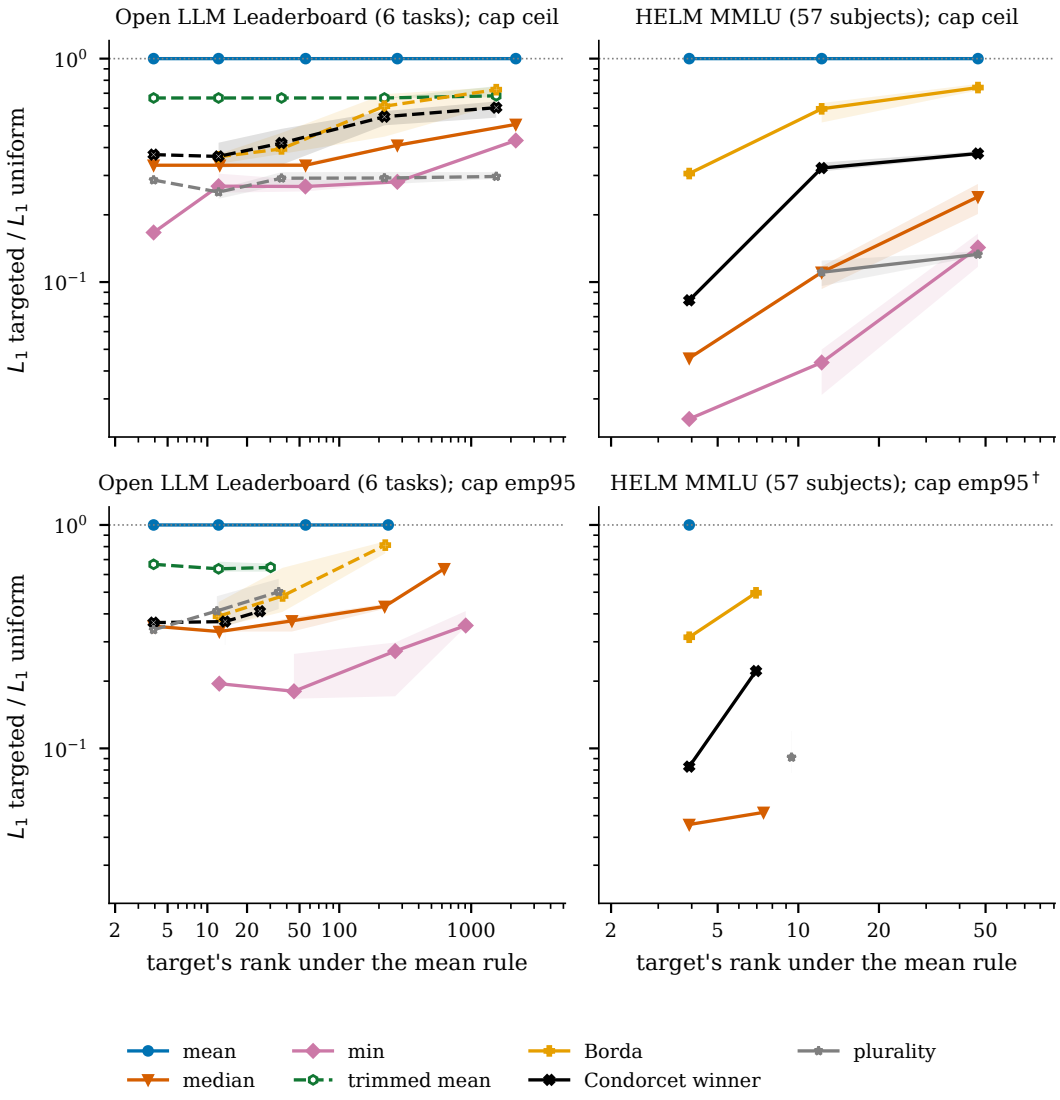


Figure 4: Ratio of the L_1 of the cheapest targeted plan to the L_1 of the uniform plan, median per rank bin over feasible targets that are not already first (95% bootstrap CI). The ratio is at most 1 by optimality and exactly 1 for the mean. Rows: ceiling and empirical cap.

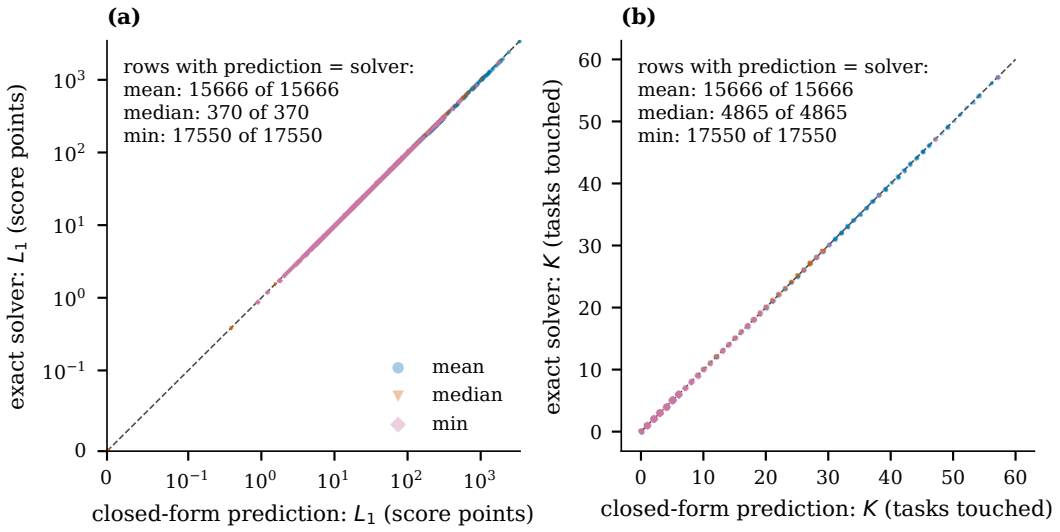


Figure 5: Closed-form prediction against exact solver value for the mean, median and min, one point per (dataset, cap, rule, target); (a) L_1, (b) K (jittered). Dashed: identity. Annotations: number of finite rows, number equal, and rows where both are infeasible (not drawn). Median L_1 has a prediction only for an odd number of tasks (HELM).

Table 3: Full version of Table 1, OLLB, cap `emp95`: rank 2, ranks 2–21 and rank >21, with K, L_1, L_∞ [95% CI] over feasible targets and infeasible counts. † rules run on the 57-target sample.

rule	targets	K	L_1	L_∞	infeas.
Open LLM Leaderboard (emp95)					
mean	rank 2	1.0	4.8	0.79	0/1
	ranks 2–21	2.0 [2.0, 2.0]	29.2 [26.5, 31.2]	5.5 [4.9, 6.0]	0/20
	rank >21	4.0 [4.0, 5.0]	66.2 [65.0, 68.1]	23.4 [22.2, 25.4]	4160/4475
geometric mean	rank 2	1.0	2.9	0.96	0/1
	ranks 2–21	4.0 [2.5, 4.0]	35.6 [21.6, 39.9]	8.6 [4.8, 10.1]	2/20
	rank >21	5.0 [5.0, 5.0]	65.2 [60.8, 69.7]	22.6 [20.0, 26.9]	4382/4475
harmonic mean	rank 2	1.0	2.0	1.0	0/1
	ranks 2–21	3.0 [1.0, 5.0]	19.8 [2.0, 45.2]	5.8 [1.0, 15.6]	9/20
	rank >21	4.0 [4.0, 5.0]	57.0 [47.8, 62.6]	19.0 [14.2, 21.1]	4409/4475
median	rank 2	1.0	9.6	4.8	0/1
	ranks 2–21	1.0 [1.0, 1.0]	3.3 [1.0, 8.5]	1.6 [0.51, 4.3]	2/20
	rank >21	2.0 [2.0, 3.0]	20.7 [18.7, 22.3]	12.8 [11.6, 14.9]	4352/4475
min	rank 2	1.0	3.7	3.7	0/1
	ranks 2–21	1.0 [1.0, 2.0]	3.1 [0.89, 6.2]	3.1 [0.89, 5.0]	13/20
	rank >21	2.0 [2.0, 2.0]	6.0 [4.8, 7.2]	3.9 [3.6, 5.0]	4417/4475
Borda†	rank 2	1.0	1.7	0.68	0/1
	ranks 2–21	2.0 [1.0, 2.0]	6.3 [4.9, 9.0]	3.6 [2.0, 5.0]	0/20
	rank >21	3.0 [2.0, 4.0]	22.2 [12.4, 32.2]	7.1 [5.7, 9.2]	14/37
Condorcet winner†	rank 2	3.0	6.7	2.7	0/1
	ranks 2–21	3.0 [2.5, 3.0]	10.9 [8.3, 12.5]	5.4 [3.7, 8.4]	8/20
	rank >21	4.0	18.0	10.1	34/37
plurality†	rank 2	2.0	3.5	2.2	0/1
	ranks 2–21	2.0 [2.0, 2.0]	16.7 [15.4, 19.6]	11.8 [10.6, 12.9]	4/20
	rank >21	2.0 [2.0, 2.0]	27.5 [19.6, 33.4]	17.8 [10.4, 19.0]	24/37
trimmed mean†	rank 2	–	4.6	1.2	0/1
	ranks 2–21	–	24.4 [21.3, 26.0]	7.3 [6.7, 9.7]	0/20
	rank >21	–	27.5 [26.1, 30.1]	9.9 [8.4, 13.3]	24/37

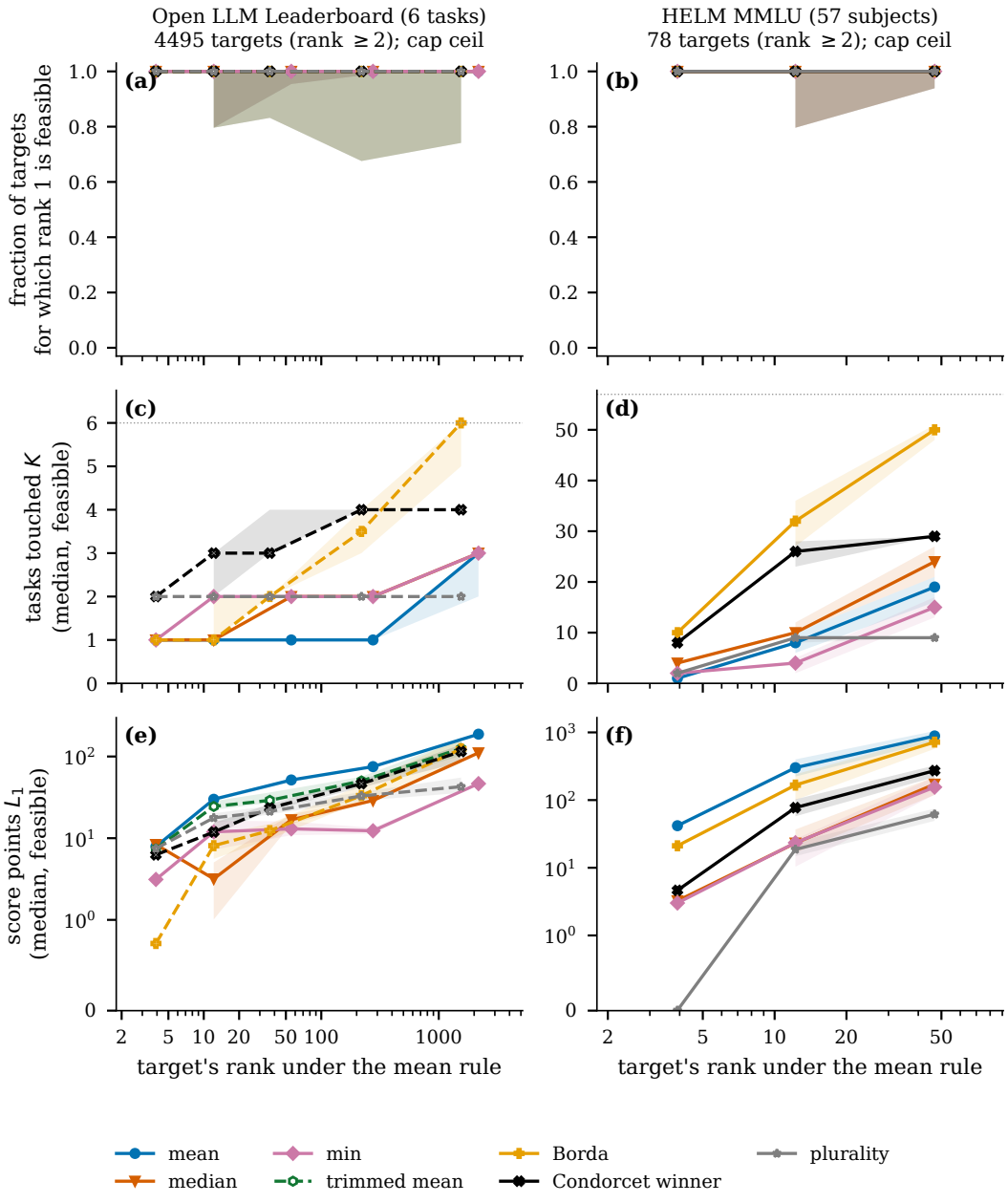


Figure 6: As Figure 1 under the permissive ceiling cap (all targets feasible), with the median K rows.

Table 4: As Table 3 for HELM, transferred cap `emp95`.

rule	targets	K	L_1	L_∞	infeas.
HELM MMLU (emp95, transferred)					
mean	rank 2	2.0	4.0	0.071	0/1
	ranks 2–21	15.0 [2.0, 55.0]	41.8 [4.0, 156]	0.73 [0.071, 2.8]	15/20
	rank >21	–	–	–	58/58
geometric mean	rank 2	0	0	0	0/1
	ranks 2–21	10.0	28.6	0.6	16/20
	rank >21	–	–	–	58/58
harmonic mean	rank 2	0	0	0	0/1
	ranks 2–21	4.0	10.9	0.32	17/20
	rank >21	–	–	–	58/58
median	rank 2	1.0	0.001	0.001	0/1
	ranks 2–21	3.5 [1.0, 7.0]	2.4 [0.38, 9.9]	1.0 [0.38, 2.3]	12/20
	rank >21	–	–	–	58/58
min	rank 2	0	0	0	0/1
	ranks 2–21	1.5	2.2	1.4	16/20
	rank >21	–	–	–	58/58
Borda	rank 2	7.0	5.7	0.67	0/1
	ranks 2–21	28.0 [7.0, 37.0]	45.2 [5.7, 59.5]	1.9 [0.67, 2.0]	13/20
	rank >21	–	–	–	58/58
Condorcet winner	rank 2	1.0	0.002	0.001	0/1
	ranks 2–21	17.0 [7.0, 19.0]	16.5 [0.89, 24.5]	1.6 [0.37, 1.8]	13/20
	rank >21	–	–	–	58/58
plurality	rank 2	4.0	1.4	0.5	0/1
	ranks 2–21	8.0 [4.5, 11.0]	6.5 [2.5, 17.1]	1.4 [0.76, 2.5]	8/20
	rank >21	–	–	–	58/58

Table 5: As Table 3 for OLLB under the ceiling cap.

rule	targets	K	L_1	L_∞	infeas.
Open LLM Leaderboard (ceil)					
mean	rank 2	1.0	4.8	0.79	0/1
	ranks 2–21	1.0 [1.0, 1.0]	29.2 [26.5, 31.2]	4.9 [4.4, 5.2]	0/20
	rank >21	2.0 [2.0, 2.0]	181 [178, 183]	30.1 [29.7, 30.4]	0/4475
geometric mean	rank 2	1.0	2.9	0.96	0/1
	ranks 2–21	1.0 [1.0, 1.0]	21.9 [18.8, 23.8]	6.1 [5.0, 6.7]	0/20
	rank >21	3.0 [3.0, 3.0]	154 [151, 155]	27.8 [27.6, 28.0]	0/4475
harmonic mean	rank 2	1.0	2.0	1.0	0/1
	ranks 2–21	1.0 [1.0, 1.0]	18.0 [13.3, 20.8]	6.7 [5.0, 8.1]	0/20
	rank >21	3.0 [3.0, 3.0]	125 [124, 127]	24.9 [24.7, 25.2]	0/4475
median	rank 2	1.0	9.6	4.8	0/1
	ranks 2–21	1.0 [1.0, 1.0]	3.6 [1.2, 9.1]	1.8 [0.58, 4.6]	0/20
	rank >21	3.0 [3.0, 3.0]	106 [105, 108]	36.2 [35.8, 36.5]	0/4475
min	rank 2	1.0	3.7	3.7	0/1
	ranks 2–21	2.0 [2.0, 2.0]	10.4 [6.3, 15.2]	6.9 [4.8, 8.7]	0/20
	rank >21	3.0 [3.0, 3.0]	43.3 [42.6, 44.2]	17.9 [17.8, 18.1]	0/4475
Borda†	rank 2	1.0	1.7	0.68	0/1
	ranks 2–21	1.0 [1.0, 2.0]	6.3 [4.8, 8.8]	3.6 [2.0, 4.7]	0/20
	rank >21	3.0 [2.0, 4.0]	24.5 [15.0, 44.9]	7.6 [5.8, 15.7]	0/37
Condorcet winner†	rank 2	3.0	6.7	2.7	0/1
	ranks 2–21	3.0 [2.0, 3.0]	11.4 [10.0, 14.4]	5.7 [4.6, 6.8]	0/20
	rank >21	4.0 [4.0, 4.0]	39.9 [26.2, 54.2]	12.9 [10.9, 17.8]	0/37
plurality†	rank 2	2.0	3.5	2.2	0/1
	ranks 2–21	2.0 [2.0, 2.0]	16.4 [14.7, 19.1]	11.0 [8.8, 11.8]	0/20
	rank >21	2.0 [2.0, 2.0]	29.5 [24.8, 35.0]	16.9 [15.2, 19.7]	0/37
trimmed mean†	rank 2	–	4.6	1.2	0/1
	ranks 2–21	–	24.4 [21.3, 26.0]	6.1 [5.3, 6.5]	0/20
	rank >21	–	45.9 [36.3, 56.5]	11.5 [9.1, 13.7]	0/37

Table 6: As Table 3 for HELM under the ceiling cap.

rule	targets	K	L_1	L_∞	infeas.
HELM MMLU (ceil)					
mean	rank 2	1.0	4.0	0.071	0/1
	ranks 2–21	7.0 [4.5, 9.0]	278 [168, 351]	5.0 [3.0, 6.3]	0/20
	rank >21	19.5 [16.0, 21.0]	914 [753, 1048]	17.1 [14.3, 20.0]	0/58
geometric mean	rank 2	0	0	0	0/1
	ranks 2–21	7.0 [4.0, 9.0]	258 [156, 327]	5.3 [3.5, 6.9]	0/20
	rank >21	19.5 [16.0, 21.0]	885 [725, 1019]	18.0 [15.4, 21.3]	0/58
harmonic mean	rank 2	0	0	0	0/1
	ranks 2–21	7.0 [4.0, 9.0]	242 [147, 306]	5.8 [4.1, 7.6]	0/20
	rank >21	20.0 [17.0, 21.5]	858 [699, 992]	18.9 [16.4, 22.7]	0/58
median	rank 2	1.0	0.001	0.001	0/1
	ranks 2–21	10.0 [6.0, 10.5]	17.8 [8.4, 34.0]	3.4 [2.0, 4.8]	0/20
	rank >21	24.5 [21.5, 27.0]	179 [123, 224]	13.6 [10.4, 15.8]	0/58
min	rank 2	0	0	0	0/1
	ranks 2–21	3.5 [2.0, 5.0]	17.0 [6.9, 25.3]	7.6 [4.4, 15.8]	0/20
	rank >21	15.5 [13.0, 17.5]	159 [120, 222]	25.0 [22.4, 27.6]	0/58
Borda	rank 2	5.0	5.6	0.67	0/1
	ranks 2–21	29.5 [19.0, 33.0]	147 [55.3, 206]	4.5 [2.1, 5.8]	0/20
	rank >21	50.0 [48.5, 51.0]	747 [588, 881]	20.1 [15.8, 23.6]	0/58
Condorcet winner	rank 2	1.0	0.002	0.001	0/1
	ranks 2–21	25.0 [18.0, 27.0]	64.8 [25.0, 94.1]	3.3 [1.8, 4.3]	0/20
	rank >21	29.0 [29.0, 29.0]	278 [220, 328]	13.6 [10.8, 15.8]	0/58
plurality	rank 2	4.0	1.4	0.5	0/1
	ranks 2–21	9.0 [7.5, 9.0]	16.0 [6.3, 21.1]	2.6 [1.4, 3.0]	0/20
	rank >21	9.0 [9.0, 9.0]	62.9 [50.1, 75.5]	8.4 [6.5, 9.9]	0/58